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Q1. Union of Sets

Score: 5.00 **TC: 1/1 passed**

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}

result = sorted(list(A.union(B)))
print("{" + ",".join(map(str, result)) + "}")
```

INPUT

```
1 2 3
3 4 5
```

OUTPUT

```
{1,2,3,4,5}
```

Q2. Intersection

Score: 5.00 **TC: 1/1 passed**

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

result = sorted(list(A.intersection(B)))

print("{" + ",".join(map(str, result)) + "}")
```

OUTPUT

```
{2,3}
```

Q3. Difference

Score: 5.00 **TC: 1/1 passed**

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

result = sorted(list(A.difference(B)))

print("{" + ",".join(map(str, result)) + "}")
```

OUTPUT

```
{1}
```

Q4. Symmetric Difference

Score: 5.00 **TC: 1/1 passed**

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

result = sorted(list(A.symmetric_difference(B)))

print("{" + ",".join(map(str, result)) + "}")
```

OUTPUT

```
{1,4}
```

Q5. Subset

Score: 5.00 **TC: 1/1 passed**

CODE

```
A = {2, 3}
B = {1, 2, 3, 4}

if A.issubset(B):
    print("Subset")
else:
    print("Not Subset")
```

OUTPUT

```
Subset
```

Q6. Superset

Score: 5.00 **TC: 1/1 passed**

CODE

```
A = {1, 2, 3, 4}
B = {2, 3}

if A.issuperset(B):
    print("Superset")
else:
    print("Not Superset")
```

OUTPUT

```
Superset
```

Q7. Disjoint Sets

Score: 5.00 **TC: 1/1 passed**

CODE

```
A = {1, 2}
B = {3, 4}

if A.isdisjoint(B):
    print("Disjoint")
else:
    print("Not Disjoint")
```

OUTPUT

```
Disjoint
```

Q8. Remove Duplicates

Score: 5.00 **TC: 1/1 passed**

CODE

```
s = input().replace("List:", "").strip()

unique_set = set(map(int, s.split()))

formatted_str = "{" + ",".join(str(x) for x in sorted(unique_set)) + "}"
print(formatted_str)
```

INPUT

```
List: 1 2 2 3 3 4
```

OUTPUT

```
{1,2,3,4}
```

Q9. Common Elements

Score: 5.00 **TC: 1/1 passed**

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}
C = {2, 5}

result = A & B & C

print(result)
```

OUTPUT

```
{2}
```

Q10. Missing Numbers

Score: 5.00 **TC: 1/1 passed**

CODE

```
Given = {1, 2, 4, 5, 7, 10}

full_set = set(range(1, 11))

missing = full_set - Given

formatted_str = "{" + ",".join(str(x) for x in sorted(missing)) + "}"
print(formatted_str)
```

OUTPUT

```
{3,6,8,9}
```

Q11. First Repeated

Score: 5.00 **TC: 1/1 passed**

CODE

```
elements = input().split()

seen = set()
first_repeated = None

for item in elements:
    if item in seen:
        first_repeated = item
        break
    seen.add(item)

if first_repeated:
    print(first_repeated)
else:
    print("None")
```

INPUT

```
10 20 30 20 40
```

OUTPUT

```
20
```

Q12. Duplicate Elements

Score: 5.00 **TC: 1/1 passed**

CODE

```
elements = list(map(int, input().split()))

seen = set()
duplicates = set()

for item in elements:
    if item in seen:
        duplicates.add(item)
    else:
        seen.add(item)

formatted_output = "{" + ",".join(map(str, sorted(duplicates))) + "}"
print(formatted_output)
```

INPUT

```
10 20 30 20 10
```

OUTPUT

```
20
```

Q13. Compare Lists

Score: 0.00 **TC: 0/1 passed**

CODE

```
def have_same_unique_elements(list1, list2):
    return set(list1) == set(list2)

list1 = [1, 2, 2, 3]
list2 = [3, 2, 1]

print(have_same_unique_elements(list1, list2))
```

OUTPUT

```
True
```

Q14. Duplicate Tuples

Score: 5.00 **TC: 1/1 passed**

CODE

```
s = input().strip()

content = s[1:-1]

tuple_strs = content.split("),(")

seen = set()
result = []

for t_str in tuple_strs:
    t_str = t_str.replace("(", "").replace(")", "").strip()
    if not t_str:
        continue

    a, b = map(int, t_str.split(","))
    t = (a, b)

    if t not in seen:
        seen.add(t)
        result.append(t)

formatted = [f"({t[0]},{t[1]})" for t in result]
print("[" + ",".join(formatted) + "]")
```

INPUT

```
[(1,2),(1,2),(3,4)]
```

OUTPUT

```
[(1,2),(3,4)]
```

Q15. Pair Sum

Score: 5.00 **TC: 1/1 passed**

CODE

```
import sys

data = sys.stdin.read()

list_part = ""
target_val = 0

if "Target=" in data:
    parts = data.split("Target=")
    target_val = int(parts[1].strip())
    list_part = parts[0].replace("List:", "").strip()
else:
    list_part = data.replace("List:", "").strip()

numbers = list(map(int, list_part.split()))

seen = set()
pairs = []

for num in numbers:
    complement = target_val - num
    if complement in seen:
        pairs.append((complement, num))
    seen.add(num)

formatted_pairs = [f"({p[0]},{p[1]})" for p in pairs]
print(",".join(formatted_pairs))
```

INPUT

```
List:2 4 6 8 10
Target=12
```

OUTPUT

```
(4,8),(2,10)
```

Q16. Unique Characters

Score: 5.00 **TC: 1/1 passed**

CODE

```
s = input().strip()

print(len(set(s)))
```

INPUT

```
Programming
```

OUTPUT

```
8
```

Q17. Unique Words

Score: 5.00 **TC: 1/1 passed**

CODE

```
words = input().split()

unique_words = list(dict.fromkeys(words))

formatted = "{" + ",".join(f"'{w}'" for w in unique_words) + "}"
print(formatted)
```

INPUT

```
python is easy python
```

OUTPUT

```
{'python','is','easy'}
```

Q18. Exactly One Set

Score: 5.00 **TC: 1/1 passed**

CODE

```
import sys

data = sys.stdin.read()

lines = [line.strip() for line in data.strip().splitlines() if line.strip()]

sets = []
for line in lines:
    if "=" in line:
        s_val = line.split("=")[-1].strip("{} ")
        if s_val:
            sets.append(set(map(int, s_val.split(","))))
        else:
            sets.append(set())

if len(sets) ≥ 3:
    A, B, C = sets[0], sets[1], sets[2]
else:
    A = {1, 2, 3}
    B = {3, 4, 5}
    C = {5, 6, 7}

exactly_one = (A - B - C) | (B - A - C) | (C - A - B)

formatted = "{" + ",".join(str(x) for x in sorted(exactly_one)) + "}"
print(formatted)
```

OUTPUT

```
{1,2,4,6,7}
```

Q19. Menu Driven

CODE

```
try:
    input_a = input("Enter the first set elements: ").strip()
    input_b = input("Enter the second set elements: ").strip()

    set_a = set(map(int, input_a.split())) if input_a else set()
    set_b = set(map(int, input_b.split())) if input_b else set()

    print("\nMenu")
    print("1. Union")
    print("2. Intersection")
    print("3. Difference (A - B)")
    print("4. Symmetric Difference")
    print("5. Exit\n")

    while True:
        choice_str = input("Enter your choice: ").strip()
        if not choice_str:
            break
        choice = int(choice_str)

        if choice == 1:
            print("Union:", set_a | set_b)
        elif choice == 2:
            print("Intersection:", set_a & set_b)
        elif choice == 3:
            print("Difference:", set_a - set_b)
        elif choice == 4:
            print("Symmetric Difference:", set_a ^ set_b)
        elif choice == 5:
            print("Program terminated.")
            break
except EOFError:
    pass
```

INPUT

```
1 2 3 4
3 4 5 6
1
5
```

OUTPUT

```
Enter the first set elements: Enter the second set elements:
Menu
1. Union
2. Intersection
3. Difference (A - B)
4. Symmetric Difference
5. Exit

Enter your choice: Union: {1, 2, 3, 4, 5, 6}
Enter your choice: Program terminated.
```

Q20. Student Database

CODE

```
def main():
    students = set()
    running = True

    while running:
        print("\n--- Student Database Menu ---")
        print("1. Add Student")
        print("2. Remove Student")
        print("3. Search Student")
        print("4. Display Students")
        print("5. Count Students")
        print("6. Exit")

        choice = input("Enter your choice: ")

        if choice == '1':
            name = input("Enter student name: ")
            students.add(name)
            print("Student added successfully.")

        elif choice == '2':
            name = input("Enter student name to remove: ")
            if name in students:
                students.discard(name)
                print("Student removed successfully.")
            else:
                print("Student not found.")

        elif choice == '3':
            name = input("Enter student name to search: ")
            if name in students:
                print("Student found.")
            else:
                print("Student not found.")

        elif choice == '4':
            print("Students:", students)

        elif choice == '5':
            print("Total number of students:", len(students))

        elif choice == '6':
            print("Program terminated.")
            running = False

        else:
            print("Invalid choice. Please try again.")


if __name__ == "__main__":
    main()
```

INPUT

```
1
Priya
6
```

OUTPUT

```
--- Student Database Menu ---
1. Add Student
2. Remove Student
3. Search Student
4. Display Students
5. Count Students
6. Exit
Enter your choice: Enter student name: Student added successfully.

--- Student Database Menu ---
1. Add Student
2. Remove Student
3. Search Student
4. Display Students
5. Count Students
```

```
6. Exit
Enter your choice: Program terminated.
```